Department of Electronic & Telecommunication Engineering
University of Moratuwa




**EN4431 - Analog IC Design**

# Assignment 02
# Second Order Type-II Integer-N Phase Locked Loop Analysis, Design, and Simulation

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**Abstract**

This report presents the analysis, design, and simulation of a second-order Type-II integer-N phase-locked loop implemented using the SKY130 CMOS process. The PLL is designed using a PFD, charge pump, passive loop filter, current-starved ring VCO, and feedback frequency divider. According to the assignment requirement, a programmable integer divider should ideally be used to cover an output frequency range of 50 MHz to 150 MHz from a 2 MHz reference. However, due to implementation-time limitations, the final transistor-level implementation uses a fixed divide-by-64 feedback divider. Therefore, the implemented PLL is verified as a fixed-frequency integer-N PLL with an expected locked output frequency of 128 MHz. The VCO block is based on the current-starved ring VCO designed in Assignment 01. The loop filter is calculated for a target loop bandwidth of 180 kHz and a phase margin of 65 degrees.

# 1 Introduction

In an integer-N PLL, the VCO output is divided by an integer value $N$ and compared with the reference clock. When the loop is locked, the divided feedback signal has the same frequency as the reference signal.

The locked frequency relationship of an integer-N PLL is

$$f_{OUT} = N f_{REF}. \tag{1}$$

The reference frequency for this design is

$$f_{REF} = 2\,\text{MHz}. \tag{2}$$

The assignment specification expects the PLL to cover an output frequency range from 50 MHz to 150 MHz. Therefore, a complete programmable divider implementation should support

$$N_{min} = \frac{50\,\text{MHz}}{2\,\text{MHz}} = 25, \tag{3}$$

and

$$N_{max} = \frac{150\,\text{MHz}}{2\,\text{MHz}} = 75. \tag{4}$$

Thus, the ideal divider range is

$$25 \leq N \leq 75. \tag{5}$$

However in the final implementation of this work, a constant divide-by-64 feedback divider was used instead of a programmable divider. This was done due to implementation-time constraints, because designing, integrating, and verifying a programmable divider over the full range would require additional digital control logic and more simulation time. Therefore, this design is evaluated as a fixed divide-by-64 integer-N PLL.

For the implemented divider,

$$N = 64. \tag{6}$$

Hence, the expected output frequency is

$$f_{OUT} = 64 \times 2\,\text{MHz} = 128\,\text{MHz}. \tag{7}$$

Therefore, the main objective of this report is to show the complete theoretical analysis, loop-filter calculation, schematic-level implementation, and simulation verification of the implemented fixed divide-by-64 Type-II charge-pump PLL.

# 2 Design Specifications

The final PLL design specifications are summarized in Table 1.

Table 1: Final PLL design specifications

| **Parameter** | **Value** |
|---|---|
| Supply voltage | 1.8 V |
| Reference frequency | 2 MHz |
| Implemented divider ratio | $N = 64$ |
| locked output frequency | 127.9993 MHz |
| Charge-pump current | 100 $\mu$A |
| Target loop bandwidth | 180 kHz |
| Target phase margin | 65° |
| VCO sensitivity | $2.6737 \times 10^9$ rad/s/V |

# 3 Theoretical Analysis

## 3.1 Transfer Function Derivations and Analysis

### Q1.1.A PFD and CP

**(i), (ii)**

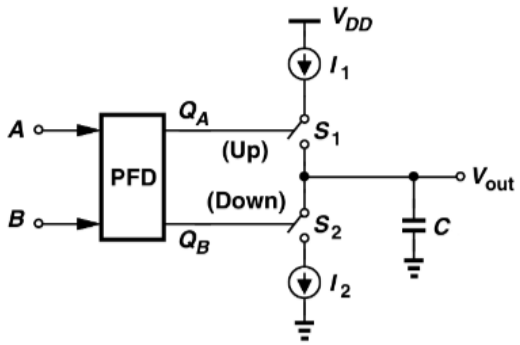



Figure 1: Cascaded PFD and CP blocks

For a phase error of $\Delta\phi$, the charge-pump pulse width is proportional to the phase error. For one reference period $T_{REF}$, the pulse width can be written as

$$t_{ON} = \frac{T_{REF}\Delta\phi}{2\pi}. \tag{8}$$

If the charge-pump current is $I_{CP}$, the average current delivered to the loop filter is

$$I_{avg} = I_{CP}\frac{t_{ON}}{T_{REF}}. \tag{9}$$

Substituting the expression for $t_{ON}$,

$$I_{avg} = I_{CP}\frac{\Delta\phi}{2\pi}. \tag{10}$$

Therefore, the PFD and charge-pump gain is

$$K_{PD} = \frac{I_{avg}}{\Delta\phi} = \frac{I_{CP}}{2\pi}. \tag{11}$$

For this design,

$$I_{CP} = 100\,\mu\text{A}. \tag{12}$$

Hence,

$$K_{PD} = \frac{100 \times 10^{-6}}{2\pi} = 15.915\,\mu\text{A/rad}. \tag{13}$$

## Q1.1.B the loop filter design

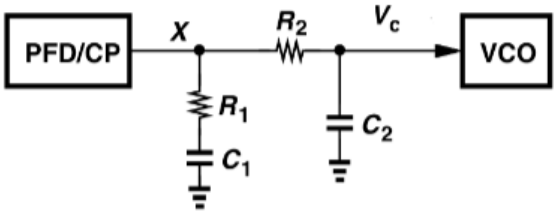



Figure 2: Second-order loop filter design

The loop filter is driven by the charge-pump current $I_{CP}(s)$, and the output of the loop filter is the VCO control voltage $V_C(s)$. Therefore, the loop-filter transfer function is defined as

$$F(s) = \frac{V_C(s)}{I_{CP}(s)}. \tag{14}$$

Let the voltage at the charge-pump output node be $V_X(s)$.

$$\frac{V_X(s) - V_C(s)}{R_2} = sC_2V_C(s). \tag{15}$$

Rearranging,

$$V_X(s) - V_C(s) = sR_2C_2V_C(s). \tag{16}$$

Hence,

$$V_X(s) = V_C(s)(1 + sR_2C_2). \tag{17}$$

Therefore,

$$V_C(s) = \frac{V_X(s)}{1 + sR_2C_2}. \tag{18}$$

Now applying KCL at node $X$, the charge-pump current is equal to the sum of the current through the $R_1 - C_1$ branch and the current through the $R_2 - C_2$ branch:

$$I_{CP}(s) = I_1(s) + I_2(s). \tag{19}$$

The impedance of the $R_1 - C_1$ branch is

$$Z_1 = R_1 + \frac{1}{sC_1}. \tag{20}$$

Therefore, the current through this branch is

$$I_1(s) = \frac{V_X(s)}{R_1 + \frac{1}{sC_1}}. \tag{21}$$

Simplifying,

$$I_1(s) = V_X(s)\frac{sC_1}{1 + sR_1C_1}. \tag{22}$$

The current through $C_2$ is

$$I_2(s) = sC_2V_C(s). \tag{23}$$

Thus, the KCL equation becomes

$$I_{CP}(s) = V_X(s)\frac{sC_1}{1 + sR_1C_1} + sC_2V_C(s). \tag{24}$$

Using

$$V_X(s) = V_C(s)(1 + sR_2C_2), \tag{25}$$

we get

$$I_{CP}(s) = V_C(s)(1 + sR_2C_2)\frac{sC_1}{1 + sR_1C_1} + sC_2V_C(s). \tag{26}$$

Factoring out $V_C(s)$,

$$I_{CP}(s) = V_C(s)\left[\frac{sC_1(1 + sR_2C_2)}{1 + sR_1C_1} + sC_2\right]. \tag{27}$$

Taking a common denominator,

$$I_{CP}(s) = V_C(s)\left[\frac{sC_1(1 + sR_2C_2) + sC_2(1 + sR_1C_1)}{1 + sR_1C_1}\right]. \tag{28}$$

Expanding the numerator,

$$sC_1(1 + sR_2C_2) + sC_2(1 + sR_1C_1) = sC_1 + s^2R_2C_1C_2 + sC_2 + s^2R_1C_1C_2. \tag{29}$$

Therefore,

$$sC_1(1 + sR_2C_2) + sC_2(1 + sR_1C_1) = s(C_1 + C_2) + s^2C_1C_2(R_1 + R_2). \tag{30}$$

Hence,

$$I_{CP}(s) = V_C(s)\left[\frac{s(C_1+C_2)+s^2C_1C_2(R_1+R_2)}{1+sR_1C_1}\right]. \tag{31}$$

Therefore, the loop-filter transfer function is

$$F(s) = \frac{V_C(s)}{I_{CP}(s)}. \tag{32}$$

Thus,

$$F(s) = \frac{1+sR_1C_1}{s(C_1+C_2)+s^2C_1C_2(R_1+R_2)}. \tag{33}$$

Factoring the denominator,

$$F(s) = \frac{1+sR_1C_1}{s(C_1+C_2)\left[1+s\frac{C_1C_2(R_1+R_2)}{C_1+C_2}\right]}. \tag{34}$$

## Q1.1.C Rearranged Loop-Filter Transfer Function

$$F(s) = \frac{1+sR_1C_1}{s(C_1+C_2)\left[1+s\frac{C_1C_2(R_1+R_2)}{C_1+C_2}\right]}. \tag{35}$$

The assignment defines

$$\tau_z = R_1C_1, \tag{36}$$

$$\tau_p = R_2C_2, \tag{37}$$

and

$$C_{tot} = C_1 + C_2. \tag{38}$$

The numerator can be directly written as

$$1 + sR_1C_1 = 1 + s\tau_z. \tag{39}$$

Also,

$$C_1 + C_2 = C_{tot}. \tag{40}$$

For the pole term, the exact expression is

$$\tau_{p,exact} = \frac{C_1C_2(R_1+R_2)}{C_1+C_2}. \tag{41}$$

Since the stabilizing capacitor is much smaller than the main loop-filter capacitor and R1 cannot be chosen arbitrarily large, because it directly sets the loop-filter zero location and affects the phase margin and practical resistor area. The pole should be high frequency hence keep R2 as desire of the pole. So in the simplified loop-filter design form used for PLL analysis, the high-frequency ripple pole is approximated as

$$\tau_p \approx R_2 C_2. \tag{42}$$

Therefore,

$$1 + s\frac{C_1 C_2 (R_1 + R_2)}{C_1 + C_2} \approx 1 + s\tau_p. \tag{43}$$

$$F(s) = \frac{1 + s\tau_z}{sC_{tot}(1 + s\tau_p)}. \tag{44}$$

Rearranging,

$$F(s) = \frac{1}{sC_{tot}} \cdot \frac{1 + s\tau_z}{1 + s\tau_p}. \tag{45}$$

## Q1.1.D Open-loop transfer function

The open-loop transfer function is obtained by multiplying the PFD/CP gain, loop-filter transfer function, VCO phase-domain transfer function, and feedback factor. Since the divider is included in the loop, the loop gain is

$$G(s) = K_{PD} F(s) \frac{K_{VCO}}{s} \frac{1}{N}. \tag{46}$$

## Q1.1.E Open-loop transfer function substitution

$$K_{PD} = \frac{I_{CP}}{2\pi}, F(s) = \frac{1}{sC_{tot}} \frac{1 + s\tau_z}{1 + s\tau_p}, \tag{47}$$

gives

$$G(s) = \frac{I_{CP} K_{VCO}}{2\pi N C_{tot}} \frac{1 + s\tau_z}{s^2 (1 + s\tau_p)}. \tag{48}$$

This is a Type-II PLL because the loop contains two integrators. One integrator is from the loop-filter capacitor, and the other is from the VCO phase integration. The loop has two poles at the origin, one stabilizing zero, and one high-frequency pole:

$$p_1 = 0, \quad p_2 = 0, \quad p_3 = -\frac{1}{\tau_p}, \quad z_1 = -\frac{1}{\tau_z}. \tag{49}$$

## Q1.1.F The loop gain of the system

Therefore, at $s = j\omega$,

$$G(j\omega) = \frac{I_{CP} K_{VCO}}{2\pi N C_{tot}} \frac{1 + j\omega\tau_z}{(j\omega)^2 (1 + j\omega\tau_p)}. \tag{50}$$

Taking the magnitude,

$$|G(j\omega)| = \frac{I_{CP} K_{VCO}}{2\pi N C_{tot} \omega^2} \sqrt{\frac{1 + (\omega\tau_z)^2}{1 + (\omega\tau_p)^2}} \tag{51}$$

The two powers of $s$ in the denominator show that the PLL is a Type-II system. One integrator comes from the loop-filter capacitor, and the other comes from the VCO phase integration.

## Q1.1.G Phase Margin of the PLL

From Q1.1.F, the loop gain of the PLL at $s = j\omega$ is,

$$G(j\omega) = \frac{I_{CP}K_{VCO}}{2\pi NC_{tot}} \frac{1 + j\omega\tau_z}{(j\omega)^2(1 + j\omega\tau_p)}. \tag{52}$$

The constant term

$$\frac{I_{CP}K_{VCO}}{2\pi NC_{tot}} \tag{53}$$

is positive, so it contributes zero phase. Therefore, the total phase of $G(j\omega)$ is determined by the factors

$$1 + j\omega\tau_z, (j\omega)^2, 1 + j\omega\tau_p. \tag{54}$$

The total open-loop phase is

$$\angle G(j\omega) = -180^\circ + \tan^{-1}(\omega\tau_z) - \tan^{-1}(\omega\tau_p). \tag{55}$$

The loop bandwidth is defined as the crossover frequency $\omega_c$, where the loop-gain magnitude is unity:

$$|G(j\omega_c)| = 1. \tag{56}$$

The phase margin is the amount of phase above $-180^\circ$ at the crossover frequency. Therefore,

$$\phi_m = 180^\circ + \angle G(j\omega_c). \tag{57}$$

Substituting the phase expression at $\omega = \omega_c$,

$$\phi_m = 180^\circ + \left[-180^\circ + \tan^{-1}(\omega_c\tau_z) - \tan^{-1}(\omega_c\tau_p)\right]. \tag{58}$$

The $180^\circ$ and $-180^\circ$ terms cancel, giving

$$\phi_m = \tan^{-1}(\omega_c\tau_z) - \tan^{-1}(\omega_c\tau_p). \tag{59}$$

## Q1.1.H: Crossover Frequency for Maximum Phase Margin

$$\frac{d}{d\omega_c}\left[\tan^{-1}(\omega_c\tau_z) - \tan^{-1}(\omega_c\tau_p)\right] = 0. \tag{60}$$

Using

$$\frac{d}{dx}\tan^{-1}(x) = \frac{1}{1 + x^2}, \tag{61}$$

$$\frac{\tau_z}{1 + (\omega_c\tau_z)^2} - \frac{\tau_p}{1 + (\omega_c\tau_p)^2} = 0. \tag{62}$$

$$\frac{\tau_z}{1 + \omega_c^2\tau_z^2} = \frac{\tau_p}{1 + \omega_c^2\tau_p^2}. \tag{63}$$

Cross-multiplying,

$$\tau_z(1 + \omega_c^2\tau_p^2) = \tau_p(1 + \omega_c^2\tau_z^2). \tag{64}$$

$$\tau_z + \omega_c^2 \tau_z \tau_p^2 = \tau_p + \omega_c^2 \tau_p \tau_z^2. \tag{65}$$

Rearranging,

$$\tau_z - \tau_p = \omega_c^2 \tau_p \tau_z^2 - \omega_c^2 \tau_z \tau_p^2. \tag{66}$$

$$\tau_z - \tau_p = \omega_c^2 \tau_z \tau_p (\tau_z - \tau_p). \tag{67}$$

Since $\tau_z \neq \tau_p$, divide both sides by $(\tau_z - \tau_p)$:

$$1 = \omega_c^2 \tau_z \tau_p. \tag{68}$$

Therefore,

$$\omega_c^2 = \frac{1}{\tau_z \tau_p}. \tag{69}$$

Hence, the crossover frequency at which the phase margin is maximum is

$$\omega_c = \frac{1}{\sqrt{\tau_z \tau_p}} \tag{70}$$

## Q1.1.I: Required Total Capacitance at Crossover Frequency

$$|G(j\omega_c)| = 1. \tag{71}$$

Therefore,

$$1 = \frac{I_{CP} K_{VCO}}{2\pi N C_{tot} \omega_c^2} \sqrt{\frac{1 + (\omega_c \tau_z)^2}{1 + (\omega_c \tau_p)^2}}. \tag{72}$$

Rearranging for $C_{tot}$,

$$C_{tot} = \frac{I_{CP} K_{VCO}}{2\pi N \omega_c^2} \sqrt{\frac{1 + (\omega_c \tau_z)^2}{1 + (\omega_c \tau_p)^2}}. \tag{73}$$

## Q1.1.J: Closed-Loop Transfer Function of the PLL

The closed-loop transfer function is defined as

$$H(s) = \frac{\phi_{out}(s)}{\phi_{ref}(s)}. \tag{74}$$

Let the forward path gain from the phase-error input to the VCO output phase be $A(s)$. The feedback divider divides the VCO output phase by $N$, so the feedback signal is

$$\phi_{fb}(s) = \frac{\phi_{out}(s)}{N}. \tag{75}$$

The phase error applied to the PFD is

$$\phi_e(s) = \phi_{ref}(s) - \phi_{fb}(s). \tag{76}$$

Therefore,

$$\phi_e(s) = \phi_{ref}(s) - \frac{\phi_{out}(s)}{N}. \tag{77}$$

The output phase is related to the phase error by the forward path gain:

$$\phi_{out}(s) = G(s)\phi_e(s). \tag{78}$$

Substituting the phase-error expression,

$$\phi_{out}(s) = G(s)\left[\phi_{ref}(s) - \frac{\phi_{out}(s)}{N}\right]. \tag{79}$$

$$\phi_{out}(s) = G(s)\phi_{ref}(s) - \frac{G(s)}{N}\phi_{out}(s). \tag{80}$$

$$\phi_{out}(s) + \frac{G(s)}{N}\phi_{out}(s) = G(s)\phi_{ref}(s). \tag{81}$$

$$\phi_{out}(s)\left[1 + \frac{G(s)}{N}\right] = G(s)\phi_{ref}(s). \tag{82}$$

$$\frac{\phi_{out}(s)}{\phi_{ref}(s)} = \frac{G(s)}{1 + \frac{G(s)}{N}}. \tag{83}$$

Thus,

$$H(s) = \frac{G(s)}{1 + \frac{G(s)}{N}}. \tag{84}$$

$$H(s) = \frac{NG(s)}{1 + G(s)}. \tag{85}$$

## 3.2 Requirements Analysis

### Q1.2.A $R_1$, $R_2$, $C_1$, and $C_2$ for $f_{OUT} = 100\,\mathrm{MHz}$

For this part, the required output frequency is

$$f_{OUT} = 100\,\mathrm{MHz}. \tag{86}$$

so the required divider ratio is

$$N = \frac{f_{OUT}}{f_{REF}}. \tag{87}$$

$$N = \frac{100\,\mathrm{MHz}}{2\,\mathrm{MHz}} = 50. \tag{88}$$

The design parameters used for the loop-filter calculation are

$$I_{CP} = 100\,\mu\mathrm{A}, \tag{89}$$

$$K_{VCO} = 2.6737 \times 10^9 \,\text{rad/s/V}, \tag{90}$$

$$f_c = 180\,\text{kHz}, \tag{91}$$

and

$$\phi_m = 65^\circ. \tag{92}$$

The crossover angular frequency is

$$\omega_c = 2\pi f_c. \tag{93}$$

$$\omega_c = 2\pi(180 \times 10^3). \tag{94}$$

$$\omega_c = 1.131 \times 10^6 \,\text{rad/s}. \tag{95}$$

From the maximum phase-margin condition,

$$\omega_c = \frac{1}{\sqrt{\tau_z \tau_p}}. \tag{96}$$

Also, for a selected phase margin,

$$\alpha = \frac{\tau_z}{\tau_p} = \frac{1 + \sin\phi_m}{1 - \sin\phi_m}. \tag{97}$$

Substituting $\phi_m = 65^\circ$,

$$\alpha = \frac{1 + \sin 65^\circ}{1 - \sin 65^\circ}. \tag{98}$$

$$\alpha = 20.346. \tag{99}$$

Therefore,

$$\sqrt{\alpha} = 4.511. \tag{100}$$

The zero time constant is

$$\tau_z = \frac{\sqrt{\alpha}}{\omega_c}. \tag{101}$$

$$\tau_z = \frac{4.511}{1.131 \times 10^6}. \tag{102}$$

$$\tau_z = 3.988\,\mu\text{s}. \tag{103}$$

The pole time constant is

$$\tau_p = \frac{1}{\sqrt{\alpha}\omega_c}. \tag{104}$$

$$\tau_p = \frac{1}{4.511(1.131 \times 10^6)}. \tag{105}$$

$$\tau_p = 0.196\,\mu\text{s}. \tag{106}$$

The required total capacitance at crossover is

$$C_{tot} = \frac{I_{CP} K_{VCO}}{2\pi N \omega_c^2} \sqrt{\frac{1 + (\omega_c \tau_z)^2}{1 + (\omega_c \tau_p)^2}}. \tag{107}$$

Substituting the values,

$$C_{tot} = \frac{(100 \times 10^{-6})(2.6737 \times 10^9)}{2\pi(50)(1.131 \times 10^6)^2} \sqrt{\frac{1 + (1.131 \times 10^6 \times 3.988 \times 10^{-6})^2}{1 + (1.131 \times 10^6 \times 0.196 \times 10^{-6})^2}}. \tag{108}$$

$$C_{tot} = 3.001\,\text{nF}. \tag{109}$$

For practical loop-filter design, choose

$$C_1 = 10 C_2. \tag{110}$$

Since

$$C_{tot} = C_1 + C_2, \tag{111}$$

we get

$$C_{tot} = 10 C_2 + C_2 = 11 C_2. \tag{112}$$

Therefore,

$$C_2 = \frac{C_{tot}}{11}. \tag{113}$$

$$C_2 = \frac{3.001\,\text{nF}}{11}. \tag{114}$$

$$C_2 = 272.8\,\text{pF}. \tag{115}$$

Then,

$$C_1 = 10 C_2. \tag{116}$$

$$C_1 = 2.728\,\text{nF}. \tag{117}$$

Now, using

$$\tau_z = R_1 C_1, \tag{118}$$

the resistor $R_1$ is

$$R_1 = \frac{\tau_z}{C_1}. \tag{119}$$

$$R_1 = \frac{3.988 \times 10^{-6}}{2.728 \times 10^{-9}}. \tag{120}$$

$$R_1 = 1.462\,\mathrm{k\Omega}. \tag{121}$$

Similarly, using

$$\tau_p = R_2 C_2, \tag{122}$$

the resistor $R_2$ is

$$R_2 = \frac{\tau_p}{C_2}. \tag{123}$$

$$R_2 = \frac{0.196 \times 10^{-6}}{272.8 \times 10^{-12}}. \tag{124}$$

$$R_2 = 718\,\Omega. \tag{125}$$

Therefore, the calculated loop-filter values for $f_{OUT} = 100\,\mathrm{MHz}$ are summarized in Table 2.

Table 2: Calculated loop-filter values for $f_{OUT} = 100\,\mathrm{MHz}$

| **Parameter** | **Calculated Value** |
|---|---|
| Divider ratio, $N$ | 50 |
| Crossover frequency, $f_c$ | 180 kHz |
| Crossover angular frequency, $\omega_c$ | $1.131 \times 10^6$ rad/s |
| Phase margin, $\phi_m$ | 65° |
| Zero time constant, $\tau_z$ | 3.988 $\mu$s |
| Pole time constant, $\tau_p$ | 0.196 $\mu$s |
| Total capacitance, $C_{tot}$ | 3.001 nF |
| Main capacitor, $C_1$ | 2.728 nF |
| Ripple capacitor, $C_2$ | 272.8 pF |
| Zero resistor, $R_1$ | 1.462 k$\Omega$ |
| Pole resistor, $R_2$ | 718 $\Omega$ |

## Q1.2.B The open-loop magnitude and phase response

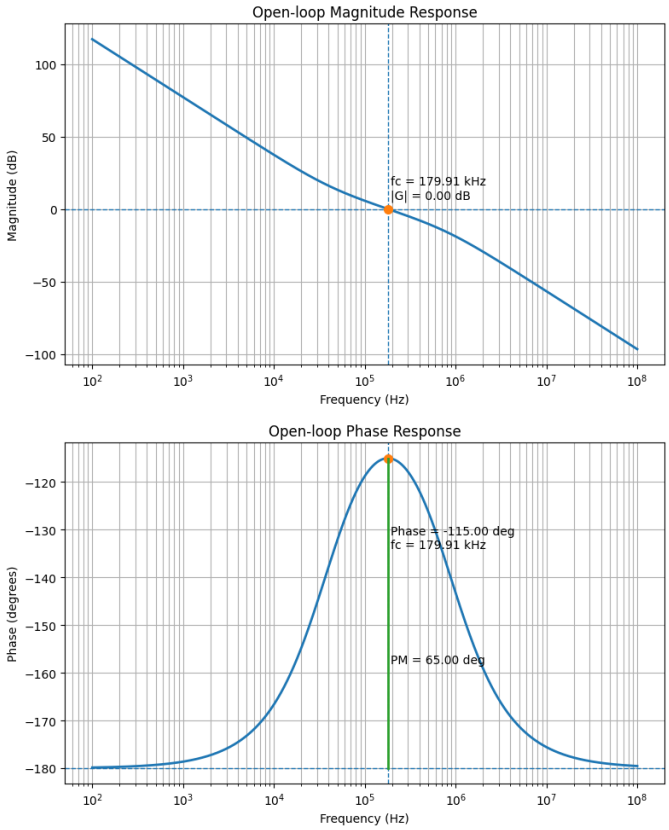



Figure 3: The open-loop magnitude and phase response

## Q1.2.C The closed-loop step response

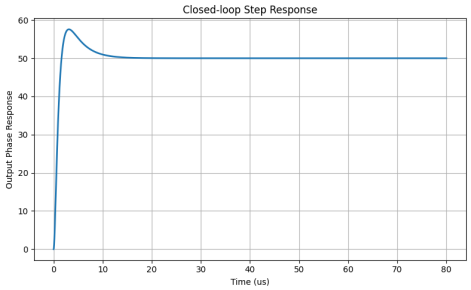



Figure 4: The closed loop step response

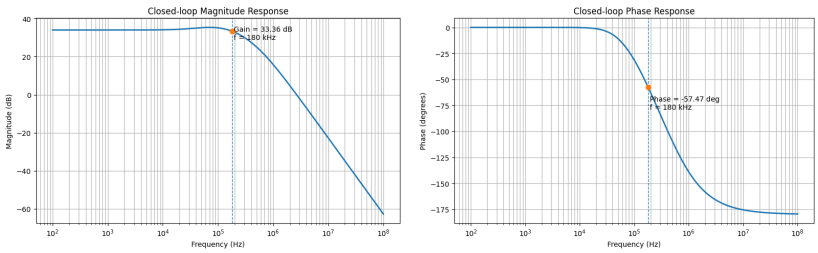



Figure 5: The closed loop magnitude and phase response

# 4 Design and Simulation

## 4.1 Blockwise Design and Simulation

### 4.1.1 PFD Implementation

The phase-frequency detector was implemented using two edge-triggered D flip-flops and reset logic. The reference clock is connected to one flip-flop, and the divided VCO output is connected to the second flip-flop. The D inputs are tied to logic high. When the reference clock edge arrives before the divided feedback edge, the UP output is asserted. When the divided feedback edge arrives before the reference edge, the DOWN output is asserted. When both outputs become high, the reset logic clears the flip-flops.

The Fig.6 and Fig.7 show the schematic and the unit test results for the PFD.

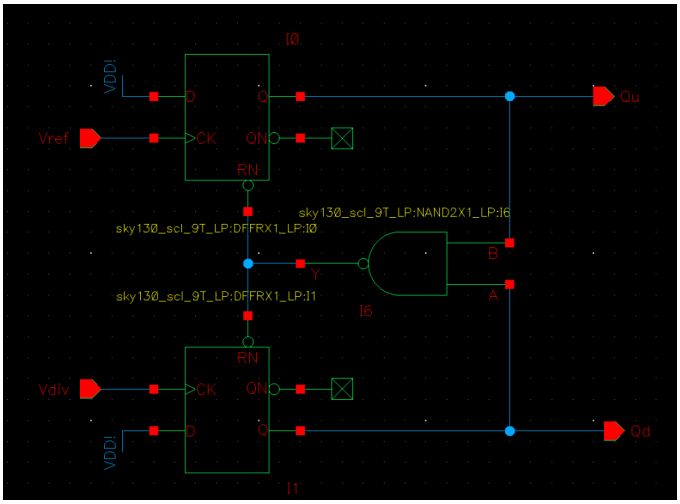



Figure 6: Schematic of the phase-frequency detector.

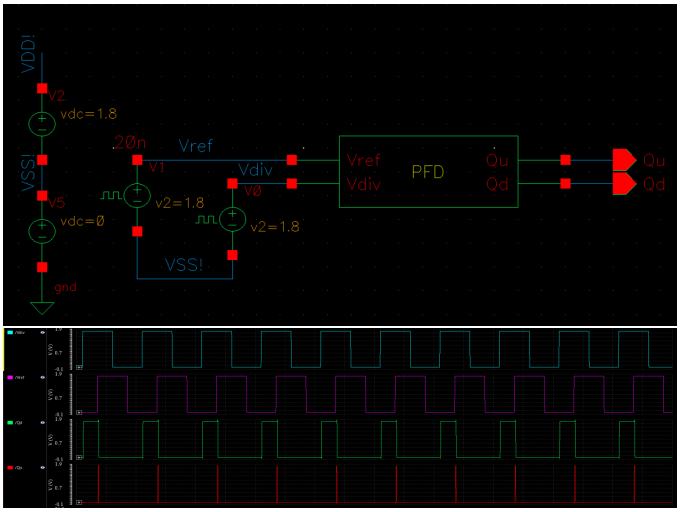



Figure 7: Testbench and results for the PFD

### 4.1.2 Charge-Pump Implementation

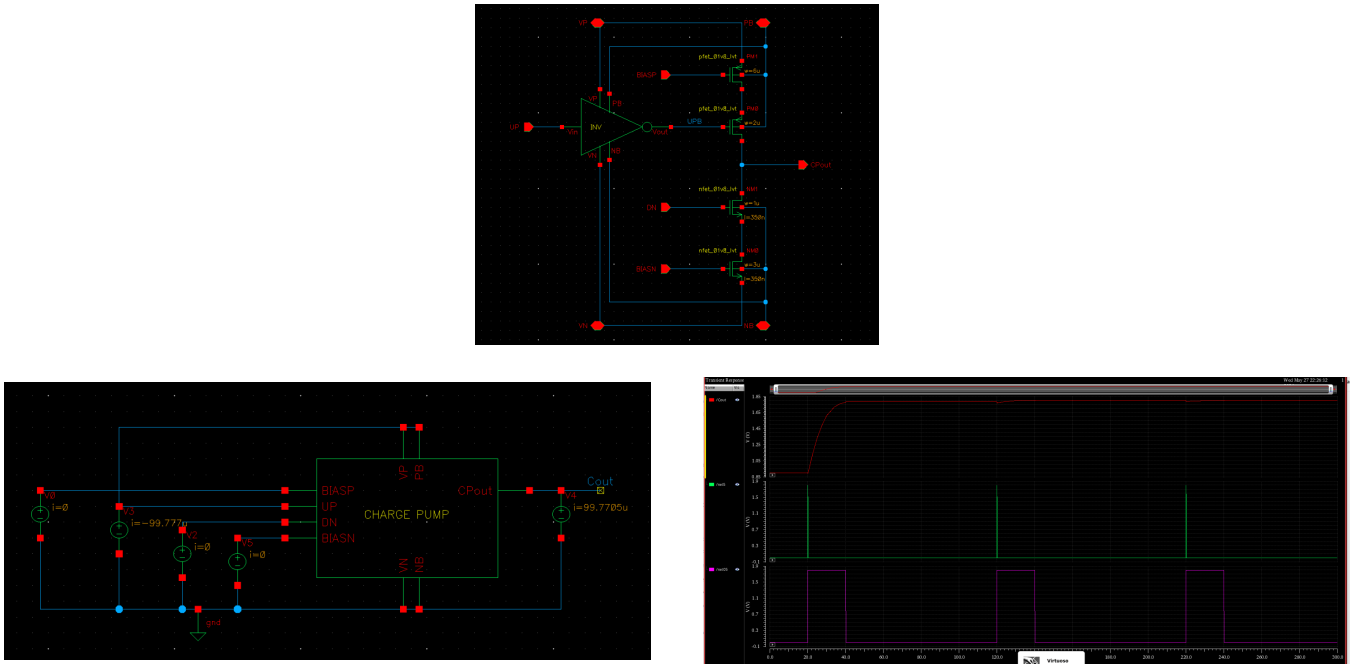



Figure 8: Schematic and testbench of the charge-pump circuit.

The charge pump converts the UP and DOWN logic signals into current pulses. The UP path charges the loop filter, increasing the VCO control voltage. The DOWN path discharges the loop filter, decreasing the VCO control voltage.

The nominal charge-pump current is

$$I_{CP} = 100\,\mu\text{A}. \tag{126}$$

Ideally, the charging and discharging currents should be equal:

$$I_{UP} = I_{DOWN} = 100\,\mu\text{A}. \tag{127}$$

Any mismatch between the UP and DOWN currents results in static phase offset and additional reference spurs. Therefore, current-source matching and switching symmetry are important for the charge-pump design. The Fig.8 shows the schematic and the test result for the CP.

### 4.1.3 Current-Starved Ring VCO Implementation

The VCO used in this PLL is the current-starved ring VCO designed and characterized in Assignment 01. It consists of cascaded inverter delay stages.

The VCO is as shown in Fig. 9 to operate at a mid-frequency of 100 MHz, with a range of 50-150 MHz, and it can comfortably operate in this PLL because the required locked frequency for the fixed divide-by-64 implementation is approximately 128 MHz, which lies well within the tuning capability of the current-starved ring oscillator.

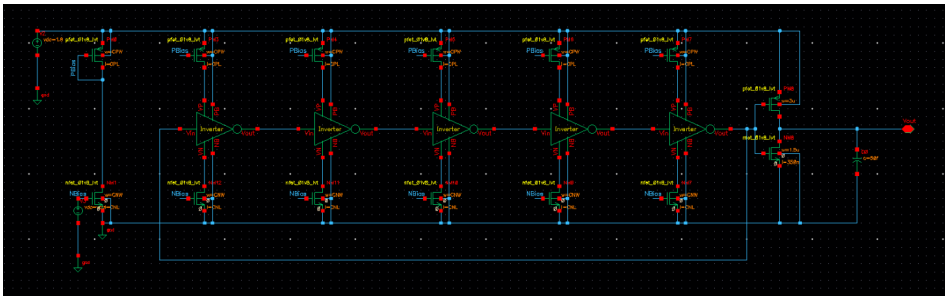


Figure 9: VCO schematic used in the PLL.

### 4.1.4 Fixed Divide-by-64 Feedback Divider

In the final implementation, a fixed divide-by-64 divider was used as shown in Fig. 10. The divider output frequency is

$$f_{DIV} = \frac{f_{OUT}}{64}. \tag{128}$$

At lock,

$$f_{DIV} = f_{REF}. \tag{129}$$

Therefore,

$$f_{OUT} = 64 f_{REF} = 128\,\text{MHz}. \tag{130}$$

Although the assignment expects a programmable divider, we used this due to the limited time constraints. So the fixed divide-by-64 divider is sufficient to verify the main closed-loop locking behaviour of the PLL. The limitation is that the output frequency is fixed for a given reference frequency.

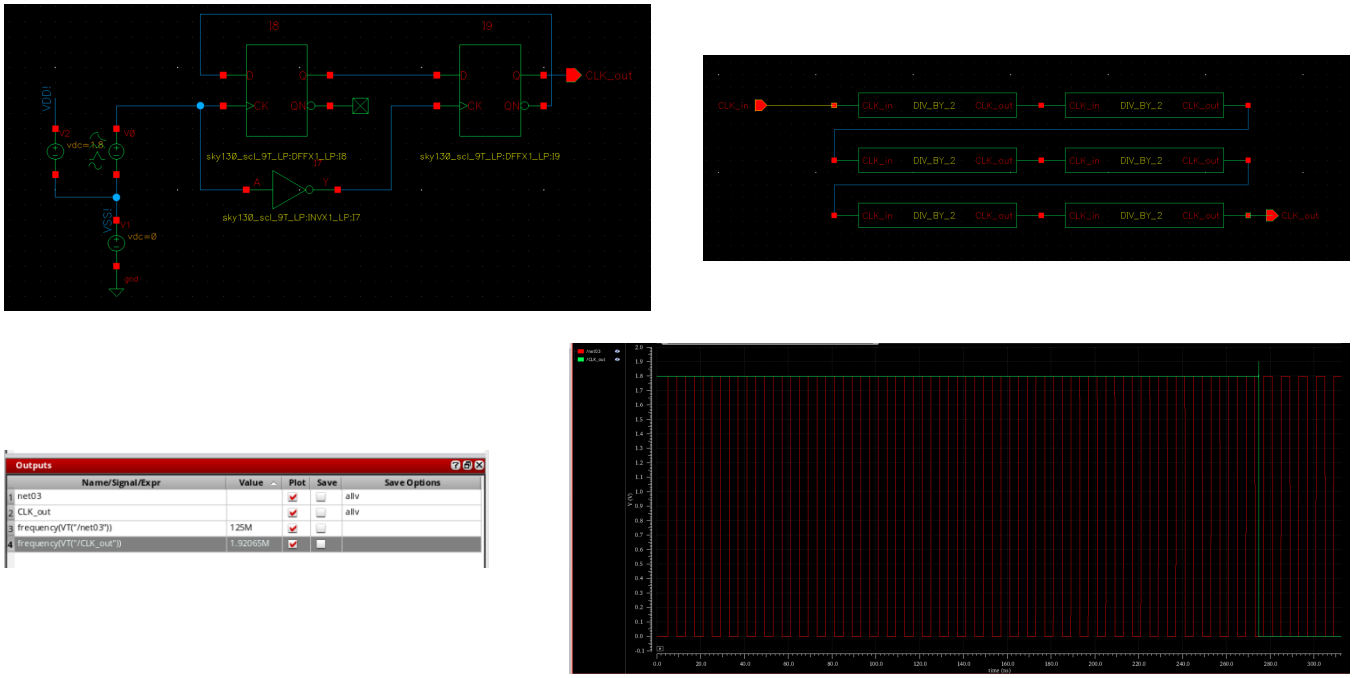



Figure 10: Simulation results of the divider circuit.

### 4.1.5 Loop filter

The loop filter was implemented as in Fig. 11 according to the calculations of R and C values above. In the real layout the capacitors will be placed as external devices since the silicon area taken for them is comparatively large for our application.

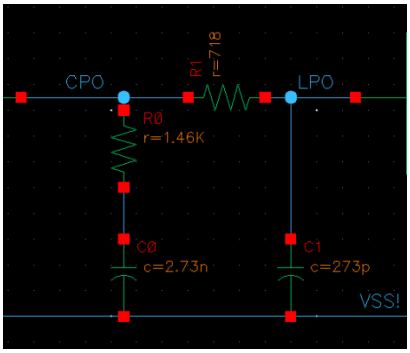



Figure 11: Loop filter design

## 4.2 Complete PLL Schematic

The complete PLL was constructed by connecting the PFD, charge pump, loop filter, VCO, and divide-by-64 divider in a feedback loop. The reference clock is applied to the PFD as below in Fig. 12.

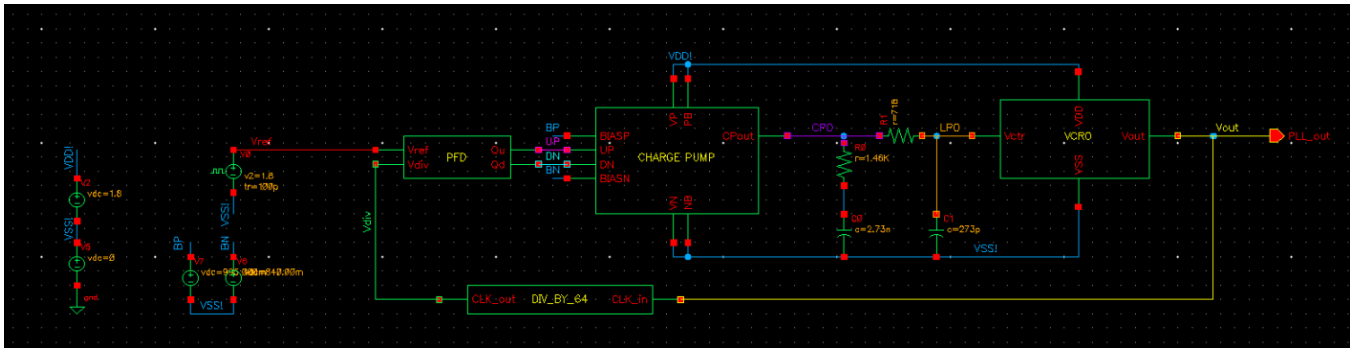



Figure 12: PLL testbench

### 4.2.1 Transient Locking Behaviour

Transient analysis was carried out to verify the locking behaviour of the complete PLL. The key signals observed in the transient simulation are the reference clock, divided feedback clock, VCO control voltage, and VCO output waveform.

During startup, the VCO frequency is not exactly equal to the locked frequency. Therefore, the PFD and charge pump adjust the loop-filter voltage until the divided feedback frequency becomes equal to the reference frequency. The locked condition is achieved at 127.995 MHz, and in the simulation, an initial condition of 0.9V was applied to the output of the charge pump to reduce the simulation time.

$$f_{DIV} = f_{REF} = 2\,\text{MHz}. \tag{131}$$

Since the divider ratio is fixed at 64,

$$f_{OUT} = 64 \times 2\,\text{MHz} = 128\,\text{MHz}. \tag{132}$$

Therefore, the expected locked VCO output frequency is approximately 128 MHz. The Figs. 13 and 14 show the pre-locking and post-locking behavior of the PLL.

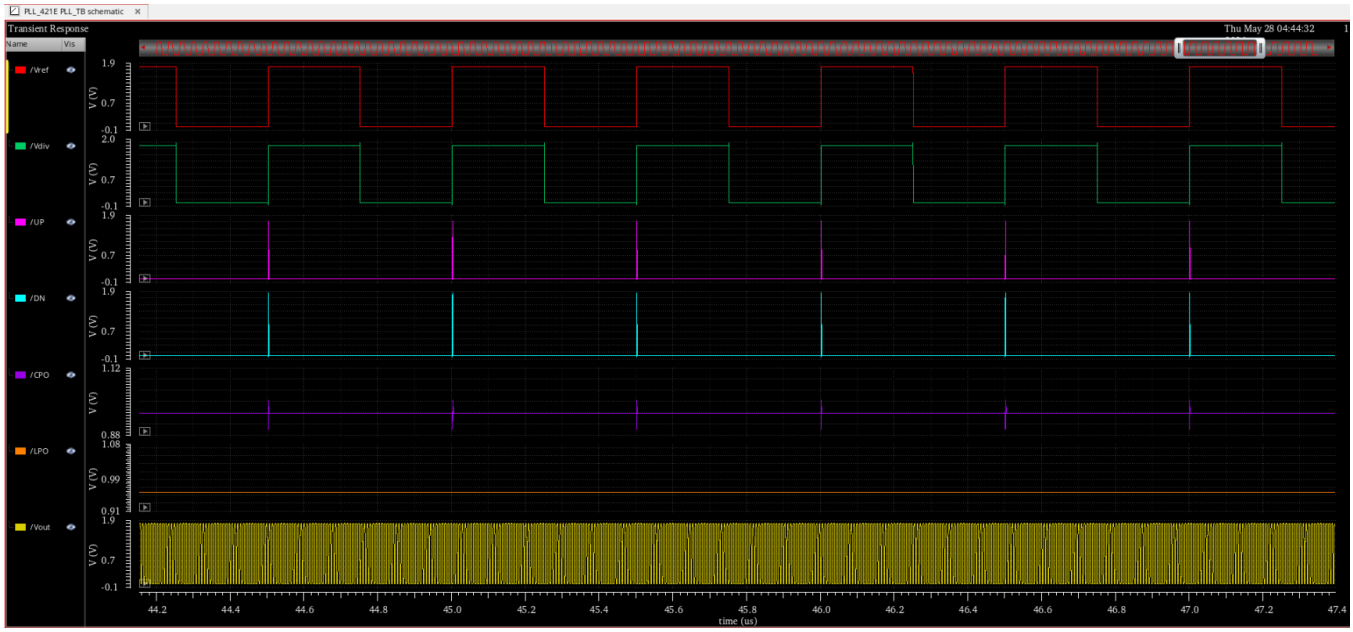



Figure 14: Locked behaviour

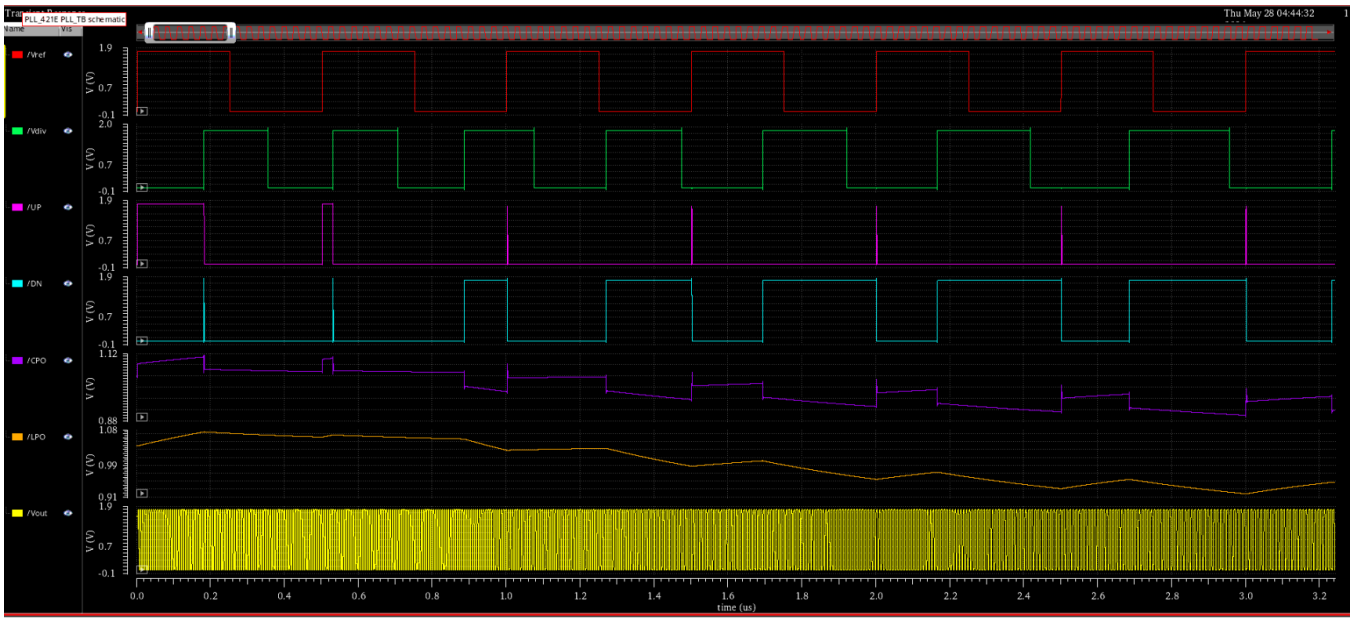



Figure 13: Locking behaviour

After the PLL reaches steady-state operation, the VCO output waveform is used for FFT analysis as shown below in Fig. 15.

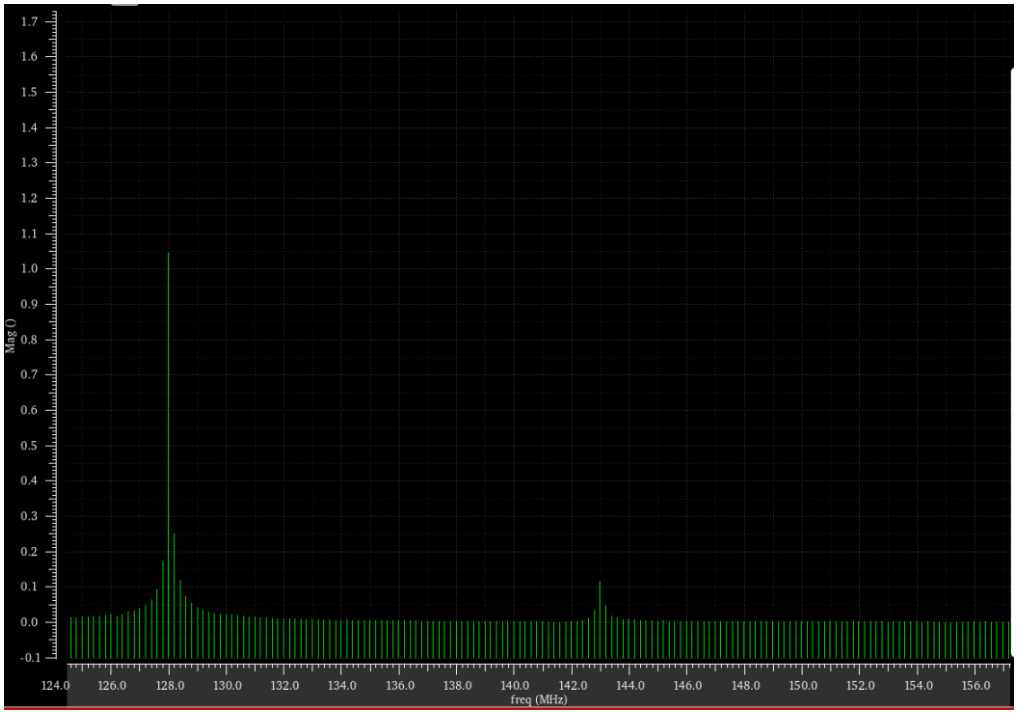



Figure 15: FFT of the locked signal

### 4.2.2 Comparison with the Assignment Requirement

The implemented fixed divide-by-64 divider PLL validates the closed-loop integer-N PLL operation at a fixed output frequency instead of demonstrating the complete programmable frequency range.

| **Feature** | **Assignment Requirement** | **Implemented Design** |
|---|---|---|
| PLL type | Type-II integer-N PLL | Type-II integer-N PLL |
| Reference frequency | 2 MHz | 2 MHz |
| Divider type | Programmable integer divider | Fixed integer divider |
| Divider range | 25 to 75 | 64 only |
| Output frequency range | 50 MHz to 150 MHz | 128 MHz only |
| Loop bandwidth | Around 180 kHz | 180 kHz |
| Minimum phase margin | At least 60 degrees | Designed for 65 degrees |
| Main limitation | None | Fixed output frequency |

Table 3: Comparison between ideal assignment requirement and implemented design

Although the fixed divider does not satisfy the full programmability requirement, it still demonstrates the essential behaviour of the PLL. The PFD, charge pump, loop filter, VCO, and divider operate together in a closed feedback loop and lock to the expected frequency determined by the divider ratio.

### 4.3 Improvement that can help achieve more performance on an Integer-N PLL

In the architecture shown in Fig. 1(a) in the tasksheet, the DSM is mainly required for fractional-N operation, where the divider modulus is dynamically varied to obtain an average fractional division ratio. If the DSM is not used, the PLL can be improved into a proper Integer-N PLL by replacing the DSM-controlled divider path with a programmable integer feedback divider.

A practical way to implement this improvement is to use a programmable modulo-(N) divider or a pulse-swallow divider. A pulse-swallow divider is specially useful when the VCO frequency is high, because the first prescaler stage operates at the highest frequency while the program and swallow counters operate at lower frequencies. This reduces the speed requirement of the main digital divider while still allowing different integer division ratios to be selected.

# References

[1] R. Jacob Baker, CMOS: Circuit Design, Layout, and Simulation, 3rd ed., Wiley, 2010.

[2] B. Razavi, Design of Analog CMOS Integrated Circuits, McGraw-Hill Education.

[3] B. Razavi, RF Microelectronics, Pearson Education.

# Appendix

```
// Generated for: spectre
// Generated on: May 28 02:28:39 2026
// Design library name: PLL_421E
// Design cell name: PLL_TB
// Design view name: schematic
simulator lang=spectre
global 0 VSS! VDD!
include "/home/aed/libs/Sky130_cadance_pdk/sky130_release_0.1.0/models/sky130.lib.spice" se

// Library name: sky130_scl_9T_LP
// Cell name: DFFRX1_LP
// View name: schematic
subckt DFFRX1_LP CK D Q QN RN inh_VDD inh_VSS
    mn36 (n25 mout n27 inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
    mn35 (n27 RN inh_VSS inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
    mn37 (n20 CKbb n25 inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
    mn26 (n20 CKb n21 inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
    mn25 (n21 D inh_VSS inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
    mn57 (QN qint inh_VSS inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
    mn55 (Q qbint inh_VSS inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
    mn45 (n35 RN inh_VSS inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
    mn46 (qbint n30 n35 inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
    mn51 (n30 CKb n40 inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
    mn50 (n40 qbint inh_VSS inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
    mn21 (CKbb CKb inh_VSS inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
    mn40 (n30 CKbb mout inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
    mn30 (mout n20 inh_VSS inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
    mn56 (qint qbint inh_VSS inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
    mn20 (CKb CK inh_VSS inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
```

```
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
    mp26 (n20 CKbb n22 inh_VDD) pfet_01v8_hvt w=(540n) l=150n as=143.1f \
        ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
    mp25 (n22 D inh_VDD inh_VDD) pfet_01v8_hvt w=(540n) l=150n as=143.1f \
        ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
    mp51 (n30 CKbb n41 inh_VDD) pfet_01v8_hvt w=(540n) l=150n as=143.1f \
        ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
    mp50 (n41 qbint inh_VDD inh_VDD) pfet_01v8_hvt w=(540n) l=150n \
        as=143.1f ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
    mp57 (QN qint inh_VDD inh_VDD) pfet_01v8_hvt w=(620n) l=150n as=164.3f \
        ad=164.3f ps=1.77u pd=1.77u m=(1)*(1)
    mp55 (Q qbint inh_VDD inh_VDD) pfet_01v8_hvt w=1.13u l=150n as=164.3f \
        ad=164.3f ps=1.77u pd=1.77u m=1
    mp20 (CKb CK inh_VDD inh_VDD) pfet_01v8_hvt w=(540n) l=150n as=143.1f \
        ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
    mp21 (CKbb CKb inh_VDD inh_VDD) pfet_01v8_hvt w=(540n) l=150n \
        as=143.1f ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
    mp36 (n26 mout inh_VDD inh_VDD) pfet_01v8_hvt w=(540n) l=150n \
        as=143.1f ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
    mp30 (mout n20 inh_VDD inh_VDD) pfet_01v8_hvt w=(540n) l=150n \
        as=143.1f ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
    mp35 (n26 RN inh_VDD inh_VDD) pfet_01v8_hvt w=(540n) l=150n as=143.1f \
        ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
    mp37 (n20 CKb n26 inh_VDD) pfet_01v8_hvt w=1.13u l=150n as=143.1f \
        ad=75.6f ps=1.61u pd=820n m=1
    mp46 (qbint n30 inh_VDD inh_VDD) pfet_01v8_hvt w=(540n) l=150n \
        as=143.1f ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
    mp40 (n30 CKb mout inh_VDD) pfet_01v8_hvt w=(540n) l=150n as=143.1f \
        ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
    mp45 (qbint RN inh_VDD inh_VDD) pfet_01v8_hvt w=(540n) l=150n \
        as=143.1f ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
    mp56 (qint qbint inh_VDD inh_VDD) pfet_01v8_hvt w=(540n) l=150n \
        as=143.1f ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
ends DFFRX1_LP
// End of subcircuit definition.

// Library name: sky130_scl_9T_LP
// Cell name: NAND2X1_LP
// View name: schematic
subckt NAND2X1_LP A B inh_VDD inh_VSS Y
    mn1 (Y B n0 inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f ad=111.3f \
        ps=1.37u pd=1.37u m=(1)*(1)
    mn0 (n0 A inh_VSS inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
    mp1 (Y B inh_VDD inh_VDD) pfet_01v8_hvt w=(620n) l=150n as=164.3f \
        ad=164.3f ps=1.77u pd=1.77u m=(1)*(1)
    mp0 (Y A inh_VDD inh_VDD) pfet_01v8_hvt w=1.13u l=150n as=164.3f \
```

```
        ad=164.3f ps=1.77u pd=1.77u m=1
ends NAND2X1_LP
// End of subcircuit definition.

// Library name: PLL_421E
// Cell name: PFD
// View name: schematic
subckt PFD Qd Qu Vdiv Vref inh_VDD inh_VSS
    I1 (Vdiv VDD! Qd net14 net11 inh_VDD inh_VSS) DFFRX1_LP
    I0 (Vref VDD! Qu net15 net11 inh_VDD inh_VSS) DFFRX1_LP
    I6 (Qd Qu inh_VDD inh_VSS net11) NAND2X1_LP
ends PFD
// End of subcircuit definition.

// Library name: PLL_421E
// Cell name: inv
// View name: schematic
subckt inv NB PB VN VP Vin Vout
    NM0 (Vout Vin VN NB) nfet_01v8_lvt w=(3u) l=350n as=795f ad=795f \
        ps=6.53u pd=6.53u m=(1)*(1)
    PM1 (Vout Vin VP PB) pfet_01v8_lvt w=(6u) l=350n as=1.59p ad=1.59p \
        ps=12.53u pd=12.53u m=(1)*(1)
ends inv
// End of subcircuit definition.

// Library name: PLL_421E
// Cell name: CHARGE_PUMP
// View name: schematic
subckt CHARGE_PUMP BIASN BIASP CPout DN NB PB UP VN VP
    NM1 (CPout DN net26 NB) nfet_01v8_lvt w=(1u) l=350n as=265f ad=265f \
        ps=2.53u pd=2.53u m=(1)*(1)
    NM0 (net26 BIASN VN NB) nfet_01v8_lvt w=(3u) l=350n as=795f ad=795f \
        ps=6.53u pd=6.53u m=(1)*(1)
    PM1 (net25 BIASP VP PB) pfet_01v8_lvt w=(6u) l=350n as=1.59p ad=1.59p \
        ps=12.53u pd=12.53u m=(1)*(1)
    PM0 (CPout UPB net25 PB) pfet_01v8_lvt w=(2u) l=350n as=530f ad=530f \
        ps=4.53u pd=4.53u m=(1)*(1)
    I0 (NB PB VN VP UP UPB) inv
ends CHARGE_PUMP
// End of subcircuit definition.

// Library name: sky130_scl_9T_LP
// Cell name: DFFX1_LP
// View name: schematic
subckt DFFX1_LP CK D Q QN inh_VDD inh_VSS
    mn26 (n20 CKb n21 inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
```

```
mn25 (n21 D inh_VSS inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
    ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
mn57 (QN qint inh_VSS inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
    ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
mn55 (Q qbint inh_VSS inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
    ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
mn50 (n35 qbint inh_VSS inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
    ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
mn51 (n30 CKb n35 inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
    ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
mn35 (n25 mout inh_VSS inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
    ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
mn36 (n20 CKbb n25 inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
    ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
mn20 (CKb CK inh_VSS inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
    ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
mn45 (qbint n30 inh_VSS inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
    ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
mn56 (qint qbint inh_VSS inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
    ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
mn21 (CKbb CKb inh_VSS inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
    ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
mn30 (mout n20 inh_VSS inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
    ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
mn40 (n30 CKbb mout inh_VSS) nfet_01v8 w=(420n) l=150n as=111.3f \
    ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
mp26 (n20 CKbb n22 inh_VDD) pfet_01v8_hvt w=(540n) l=150n as=143.1f \
    ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
mp25 (n22 D inh_VDD inh_VDD) pfet_01v8_hvt w=(540n) l=150n as=143.1f \
    ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
mp51 (n30 CKbb n36 inh_VDD) pfet_01v8_hvt w=(540n) l=150n as=143.1f \
    ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
mp50 (n36 qbint inh_VDD inh_VDD) pfet_01v8_hvt w=(540n) l=150n \
    as=143.1f ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
mp57 (QN qint inh_VDD inh_VDD) pfet_01v8_hvt w=(620n) l=150n as=164.3f \
    ad=164.3f ps=1.77u pd=1.77u m=(1)*(1)
mp55 (Q qbint inh_VDD inh_VDD) pfet_01v8_hvt w=625n l=150n as=164.3f \
    ad=164.3f ps=1.77u pd=1.77u m=1
mp35 (n26 mout inh_VDD inh_VDD) pfet_01v8_hvt w=(540n) l=150n \
    as=143.1f ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
mp36 (n20 CKb n26 inh_VDD) pfet_01v8_hvt w=625n l=150n as=75.6f \
    ad=143.1f ps=820n pd=1.61u m=1
mp20 (CKb CK inh_VDD inh_VDD) pfet_01v8_hvt w=(540n) l=150n as=143.1f \
    ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
mp45 (qbint n30 inh_VDD inh_VDD) pfet_01v8_hvt w=(540n) l=150n \
    as=143.1f ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
mp56 (qint qbint inh_VDD inh_VDD) pfet_01v8_hvt w=(540n) l=150n \
```

```
        as=143.1f ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
    mp21 (CKbb CKb inh_VDD inh_VDD) pfet_01v8_hvt w=(540n) l=150n \
        as=143.1f ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
    mp40 (n30 CKb mout inh_VDD) pfet_01v8_hvt w=(540n) l=150n as=143.1f \
        ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
    mp30 (mout n20 inh_VDD inh_VDD) pfet_01v8_hvt w=(540n) l=150n \
        as=143.1f ad=143.1f ps=1.61u pd=1.61u m=(1)*(1)
ends DFFX1_LP
// End of subcircuit definition.

// Library name: sky130_scl_9T_LP
// Cell name: INVX1_LP
// View name: schematic
subckt INVX1_LP A inh_VDD inh_VSS Y
    mn0 (Y A inh_VSS inh_VSS) nfet_01v8 w=760n l=150n as=111.3f ad=111.3f \
        ps=1.37u pd=1.37u m=1
    mp0 (Y A inh_VDD inh_VDD) pfet_01v8_hvt w=(620n) l=150n as=164.3f \
        ad=164.3f ps=1.77u pd=1.77u m=(1)*(1)
ends INVX1_LP
// End of subcircuit definition.

// Library name: PLL_421E
// Cell name: DIV_BY_2
// View name: schematic
subckt DIV_BY_2 CLK_in CLK_out inh_VDD inh_VSS
    I8 (CLK_in net08 net07 net15 inh_VDD inh_VSS) DFFX1_LP
    I9 (net05 net07 CLK_out net08 inh_VDD inh_VSS) DFFX1_LP
    I7 (CLK_in inh_VDD inh_VSS net05) INVX1_LP
ends DIV_BY_2
// End of subcircuit definition.

// Library name: PLL_421E
// Cell name: DIV_BY_48
// View name: schematic
subckt DIV_BY_48 CLK_in CLK_out inh_VDD inh_VSS
    I16 (net08 CLK_out inh_VDD inh_VSS) DIV_BY_2
    I15 (net05 net08 inh_VDD inh_VSS) DIV_BY_2
    I14 (net09 net05 inh_VDD inh_VSS) DIV_BY_2
    I13 (net06 net09 inh_VDD inh_VSS) DIV_BY_2
    I12 (net07 net06 inh_VDD inh_VSS) DIV_BY_2
    I11 (CLK_in net07 inh_VDD inh_VSS) DIV_BY_2
ends DIV_BY_48
// End of subcircuit definition.

// Library name: PLL_421E
// Cell name: inverter
// View name: schematic
```

```
subckt inverter NB PB VN VP Vin Vout
    PM0 (Vout Vin VP PB) pfet_01v8_lvt w=(6u) l=350n as=1.59p ad=1.59p \
        ps=12.53u pd=12.53u m=(1)*(1)
    NM0 (Vout Vin VN NB) nfet_01v8_lvt w=(3u) l=350n as=795f ad=795f \
        ps=6.53u pd=6.53u m=(1)*(1)
ends inverter
// End of subcircuit definition.

// Library name: PLL_421E
// Cell name: VCRO
// View name: schematic
subckt VCRO VDD VSS NBias Vout
    NM17 (net021 NBias VSS VSS) nfet_01v8_lvt w=(750n) l=350n as=198.75f \
        ad=198.75f ps=2.03u pd=2.03u m=(1)*(1)
    NM16 (net024 NBias VSS VSS) nfet_01v8_lvt w=(750n) l=350n as=198.75f \
        ad=198.75f ps=2.03u pd=2.03u m=(1)*(1)
    NM15 (net027 NBias VSS VSS) nfet_01v8_lvt w=(750n) l=350n as=198.75f \
        ad=198.75f ps=2.03u pd=2.03u m=(1)*(1)
    NM14 (net13 NBias VSS VSS) nfet_01v8_lvt w=(750n) l=350n as=198.75f \
        ad=198.75f ps=2.03u pd=2.03u m=(1)*(1)
    NM13 (net14 NBias VSS VSS) nfet_01v8_lvt w=(750n) l=350n as=198.75f \
        ad=198.75f ps=2.03u pd=2.03u m=(1)*(1)
    NM8 (Vout net015 VSS VSS) nfet_01v8_lvt w=(840n) l=150n as=111.3f \
        ad=111.3f ps=1.37u pd=1.37u m=(1)*(1)
    NM1 (PBias NBias VSS VSS) nfet_01v8_lvt w=(750n) l=350n as=198.75f \
        ad=198.75f ps=2.03u pd=2.03u m=(1)*(1)
    I10 (VSS VDD net021 net020 net022 net015) inverter
    I9 (VSS VDD net024 net025 net023 net022) inverter
    I8 (VSS VDD net027 net026 net028 net023) inverter
    I6 (VSS VDD net14 net16 net015 net029) inverter
    I7 (VSS VDD net13 net15 net029 net028) inverter
    PM13 (net020 PBias VDD VDD) pfet_01v8_lvt w=(1.5u) l=350n as=397.5f \
        ad=397.5f ps=3.53u pd=3.53u m=(1)*(1)
    PM12 (net025 PBias VDD VDD) pfet_01v8_lvt w=(1.5u) l=350n as=397.5f \
        ad=397.5f ps=3.53u pd=3.53u m=(1)*(1)
    PM11 (net026 PBias VDD VDD) pfet_01v8_lvt w=(1.5u) l=350n as=397.5f \
        ad=397.5f ps=3.53u pd=3.53u m=(1)*(1)
    PM10 (net15 PBias VDD VDD) pfet_01v8_lvt w=(1.5u) l=350n as=397.5f \
        ad=397.5f ps=3.53u pd=3.53u m=(1)*(1)
    PM9 (net16 PBias VDD VDD) pfet_01v8_lvt w=(1.5u) l=350n as=397.5f \
        ad=397.5f ps=3.53u pd=3.53u m=(1)*(1)
    PM8 (Vout net015 VDD VDD) pfet_01v8_lvt w=(5.2u) l=350n as=145.75f \
        ad=145.75f ps=1.63u pd=1.63u m=(1)*(1)
    PM0 (PBias PBias VDD VDD) pfet_01v8_lvt w=(1.5u) l=350n as=397.5f \
        ad=397.5f ps=3.53u pd=3.53u m=(1)*(1)
    C0 (Vout VSS) capacitor c=50f
ends VCRO
```

```
// End of subcircuit definition.

// Library name: PLL_421E
// Cell name: PLL_TB
// View name: schematic
I0 (DN UP Vdiv Vref VDD! VSS!) PFD
V0 (Vref VSS!) vsource type=pulse val0=0 val1=1.8 period=500n delay=2n \
        rise=100p fall=100p width=250n
V7 (BP VSS!) vsource dc=995.000m type=dc
V2 (VDD! VSS!) vsource dc=1.8 type=dc
V6 (BN VSS!) vsource dc=840.00m type=dc
V5 (VSS! 0) vsource dc=0 type=dc
I8 (BN BP CPO DN VSS! VDD! UP VSS! VDD!) CHARGE_PUMP
I11 (Vout Vdiv VDD! VSS!) DIV_BY_48
I10 (VDD! VSS! LPO Vout) VCRO
C1 (LPO VSS!) capacitor c=273p
C0 (net013 VSS!) capacitor c=2.73n
R1 (CPO LPO) resistor r=718
R0 (CPO net013) resistor r=1.46K
ic CPO=0.9
simulatorOptions options reltol=1e-3 vabstol=1e-6 iabstol=1e-12 temp=27 \
    tnom=27 scalem=1.0 scale=1.0 gmin=1e-12 rforce=1 maxnotes=5 maxwarns=5 \
    digits=5 cols=80 pivrel=1e-3 sensfile="../psf/sens.output" \
    checklimitdest=psf
tran tran stop=20u errpreset=moderate step=500n write="spectre.ic" \
    writefinal="spectre.fc" annotate=status maxiters=5
finalTimeOP info what=oppoint where=rawfile
modelParameter info what=models where=rawfile
element info what=inst where=rawfile
outputParameter info what=output where=rawfile
designParamVals info what=parameters where=rawfile
primitives info what=primitives where=rawfile
subckts info what=subckts where=rawfile
saveOptions options save=allpub
```